

PriRAG: Privacy-Preserving Inference for Retrieval-Augmented Large Language Models

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Abstract—Retrieval-Augmented Generation (RAG) services enhance LLMs by dynamically retrieving relevant information from external knowledge sources, addressing their limitations in handling up-to-date or domain-specific knowledge. Under the constraints of the cloud-native LLM service paradigm, many existing RAG systems are typically deployed in cloud environments and rely on retrieving raw information from local datasets of users or third-party knowledge providers to enhance generation. However, these cloud-based RAG services often neglect critical privacy and intellectual property concerns of the knowledge base, potentially leaking sensitive information during deployment or inference. To address these risks, we propose a privacy-preserving framework (PriRAG) via proxy knowledge for cloud-based RAG. The core idea is to compress the raw data of the retrieved information into proxy knowledge representations with carefully-designed privacy preservation mechanisms. We first propose a knowledge construction mechanism that aligns proxy representations with the semantic space of LLMs to preserve utility. Then we develop a query-centric knowledge representation method with a tailored training strategy to selectively exclude the irrelevant content, further reducing the exposure of privacy-sensitive content. We further formulate a differential privacy enhanced collaborative inference strategy to secure the whole framework with certifiable guarantees. Extensive experiments on 7 public benchmarks and 4 LLMs show that PriRAG achieves a favorable balance between utility and privacy, significantly reducing privacy risks while maintaining high-quality generation.

Index Terms—Privacy preservation, retrieval-augmented generation, cloud-based service.

I. INTRODUCTION

IN RECENT years, large language models (LLMs) have seen extensive adoption across various domains. Given the

substantial computational resources required for their deployment, cloud-based LLM services that offer accessible interfaces to end users [1], [2] have emerged as the dominant paradigm. However, the effectiveness of LLMs is often constrained by the static nature of their training data, which may lack up-to-date or domain-specific knowledge. To address this limitation, RAG has been proposed as a powerful paradigm, where external knowledge sources are dynamically queried to retrieve relevant information that supplements the LLM's responses [3]. This paradigm significantly enhances the relevance and factual accuracy of model outputs. Consequently, the combination of local retrieval database or third-party knowledge source with LLMs deployed in cloud (i.e., cloud-based RAG) has become a mainstream solution for enhancing LLM utility in real-world scenarios. Despite its effectiveness, cloud-based RAG services raise critical privacy and intellectual property concerns [4], [5]. When retrieving context from external sources such as proprietary databases and private organizational knowledge, the retrieved knowledge may contain confidential information or legally protected material. In traditional RAG systems, the retrieved content is directly shared in plaintext with the cloud service provider, posing significant privacy risks. For instance, in healthcare applications, cloud-based medical LLM applications need to process retrieved sensitive patient records from hospital local databases [6], [7], creating potential compliance violations with privacy regulations like GDPR [8]. Even worse, cloud-based RAG service deployments frequently persist retrieved knowledge, caching it on remote servers or recycling them as training data for subsequent model fine-tuning. This creates permanent copies of potentially sensitive information under third-party control, with demonstrated cases where proprietary content has been inadvertently memorized and reproduced by cloud-based LLMs [9], [10], leading to further dissemination across the Internet [11]. Therefore, protecting the privacy of retrieved knowledge becomes imperative.

However, the mainstream RAG research usually prioritizes improving the effectiveness and efficiency of retrieval and generation process [12], while overlooking the privacy risks of sharing retrieved knowledge. Several recent studies explore ways to enhance privacy in RAG systems, which can be broadly classified into two categories. The first involves identifying sensitive content and rewriting the retrieved passages before sharing [13], [14]. Although these methods avoid directly transmitting the original data, the rewritten content is still shared in

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plaintext, making it susceptible to potential information leakage at the semantic level. The second class of methods employ local differential privacy techniques (LDP) [15], [16], injecting noise into the embedding sequence of the retrieved knowledge. However, since retrieved knowledge documents are typically long, it usually needs to incorporate high-intensity perturbation noise into LLM inference for privacy protection, which can significantly degrade overall performance. Therefore, safeguarding privacy in the context of cloud-based RAG remains an open and challenging research problem.

In general, retrieved knowledge documents are usually long-form and contain rich and detailed information, much of which may be redundant for addressing the target query [17]. Such redundancy not only increases the risk of privacy leakage but also introduces irrelevant noise that can adversely impact the inference process. Therefore, compressing the retrieved knowledge presents a promising solution to simultaneously mitigate privacy risks and maintain inference performance. Building upon this insight, we propose a privacy-preserving cloud-based RAG framework (PriRAG) that transforms retrieved documents into proxy knowledge representations for secure sharing with cloud-based LLMs. At the core of PriRAG is a knowledge encoder trained to generate compressed representations that are both semantically informative and privacy-preserving. To achieve this, we first introduce a knowledge construction task to optimize the encoder, which aligns the compressed knowledge representations with the semantic space of the target LLM, thereby enabling effective downstream inference. To further enhance privacy and utility, we design a query-centric knowledge representation strategy that enhances the encoder by selectively emphasizing query-relevant content while suppressing redundant information. In addition, we develop a privacy-preserving collaborative inference strategy that injects LDP noise into the proxy knowledge representations prior to transmission, which offers provable privacy guarantees. This transformation allows LLMs to leverage external knowledge without accessing the plaintext of sensitive documents.

We conduct extensive experiments across seven benchmarks and four mainstream LLMs of varying sizes. Experiments demonstrate that PriRAG maintains generation quality on par with conventional RAG methods, while significantly reducing privacy leakage. Furthermore, PriRAG exhibits strong generalization across LLMs and knowledge encoder models with diverse architectures and scales. Notably, the training of the offline knowledge encoder is highly efficient and incurs minimal computational cost in the proposed framework. In addition, due to its use of compressed contextual inputs, PriRAG can even improve online inference latency relative to standard RAG frameworks. Finally, we release a suite of optimized knowledge encoders tailored for several popular open-source large language models (e.g., the Qwen series), which can serve as foundational components in privacy-preserving, cloud-based RAG frameworks. While current closed-source model platforms primarily engage with users via unsecured plain-text interactions, we envision that our approach can contribute meaningfully to enhancing privacy protection in real-world RAG services offered by model providers. The implementation codes, aligned encoders, and datasets are released and can be accessed at <https://github.com/wanghl21/PrivacyRAG>. The main contributions of this work are shown as follows:

• We propose PriRAG, a novel framework that compresses plaintext into proxy knowledge representations for privacy-preserving sharing with cloud-based LLMs.

- We provide both theoretical privacy guarantees and empirical evaluation, demonstrating that PriRAG achieves a favorable privacy–utility trade-off, highlighting its practicality for real-world applications.
- We release a suite of knowledge encoders compatible with multiple mainstream open-source LLMs, facilitating the adoption and further development of secure RAG.

II. RELATED WORK

A. Privacy-Preserving Methods for Prompt

Most privacy-preserving methods in the field of LLMs primarily focus on protecting sensitive content during the prompting stage. Based on the type of privacy protection mechanism employed, these approaches can be broadly categorized into DP-based methods and non-DP based methods.

DP-based methods typically aim to transform raw textual inputs into embedding vectors and introduce noise to preserve privacy guarantees [16], [18], [19] or generate synthetic differentially private demonstrations for in-context learning (ICL) [20], [21]. For example, Plant et al. [18] proposed a context-aware private embedding method which disrupted embedding by combining differential privacy and adversarial learning to safeguard sensitive user information during language model training. Du et al. [16] imposed a consistency constraint on sanitizing sentence embeddings to achieve the metric LDP. Tang et al. [20] focused on the privacy protection of demonstration examples used in the prompt. Their developed algorithm generates synthetic differentially private few-shot demonstrations from the original private dataset, and uses the generated samples as demonstrations in ICL during inference. Hong et al. [21] used a few samples as demonstrations to guide a local LLM to generate private prompts. The prompt generation process was facilitated by a differentially private ensemble of in-context learning with disjoint private demonstration subsets. However, many of these methods primarily focus on protecting short queries or demonstration content, and their performance often degrades in long knowledge scenarios due to added noise or semantic inconsistencies in the privatized inputs.

Non-DP methods operate directly on plaintext inputs and include techniques such as sanitization [22], [23], ensemble prompting [24], obfuscation [25], [26], [27], and cryptographic encryption [28], [29]. For example, Kan et al. [22] proposed a local privacy-protecting module that sanitizes sensitive information before it is sent to a cloud LLM, coupled with a local privacy recovery module that reconstructs privacy in the response. While effective in hiding sensitive information, this approach may introduce semantic contradictions, potentially reducing the clarity and effectiveness of the interaction with the LLM. In another line of work, Hou et al. [28] introduced CipherGPT, which leverages homomorphic encryption and secret sharing to perform encrypted inference over encrypted inputs. Although these cryptographic approaches provide strong theoretical privacy

guarantees, they often incur substantial computational overhead. For instance, end-to-end reasoning in CipherGPT requires approximately 20 minutes per token, limiting its practicality for real-time RAG applications.

While these methods offer effective prompt-level privacy protection, they fall short in cloud-based RAG scenarios, where privacy risks primarily arise from the retrieved knowledge rather than the user query. In cloud-based RAG, the prompt sent to the server often includes extensive retrieved content that far exceeds the length and sensitivity of the original query, making knowledge-level protection essential.

B. Privacy-Enhanced Methods for RAG

Despite the growing interest in privacy-preserving RAG, existing works can be broadly categorized according to different deployment settings, which fundamentally determine the associated privacy risks and protection strategies.

1) Cloud-based LLMs with cloud-hosted data. In this setting, both the LLM and the datastore are deployed on the cloud, and the primary concern is preventing the model from leaking sensitive information contained in the database through its outputs. Existing works focus on DP-based mechanisms to mitigate such leakage. For example, Koga et al. [15] leverage sparse vector representations and DP techniques to reduce the exposure of knowledge in generated outputs. Wu et al. [30] further address this issue by introducing document-level differential privacy with per-document privacy filters, ensuring that model outputs do not reveal private knowledge. However, these approaches operate at the output level and treat privacy leakage as a property of the final generated text, without considering the risk that sensitive retrieved content may be directly processed by an untrusted cloud.

2) Local LLMs with cloud-based retrieval. In this setting, the LLM is deployed locally while the datastore resides on the cloud, and thus the primary focus is on preserving data privacy during the cloud-based retrieval process. To this end, prior works focus on secure retrieval over protected embeddings. For example, Ye et al. [31] propose conditional approximate distance-preserving symmetric encryption for retrieval over encrypted embeddings. Ming et al. [32] design a two-server protocol based on secret sharing to support secure top- k retrieval. These methods effectively protect the datastore and retrieval process; however, they rely on the assumption of local LLM deployment and therefore do not explicitly address how retrieved knowledge may be exposed during generation in cloud-based LLM inference settings.

3) Cloud-based LLMs with local private data. In this increasingly practical setting, the LLM is hosted on the cloud while private knowledge remains local, requiring knowledge to be transmitted for generation. A line of work addresses this by transforming the retrieved content before transmission. For example, Zeng et al. [13], Wang et al. [33], and Wei et al. [34] employ rewriting or masking strategies to sanitize sensitive information. Another direction is cryptographic methods, such as CipeRAG [35], which integrates cryptographic mechanisms

into the retrieval and generation process to provide privacy guarantees. However, transformation-based methods rely on accurate identification of sensitive content and may degrade utility, while cryptographic approaches often incur substantial computational overhead (e.g., $\sim 20\times$ inference latency [35]), limiting their practicality in real-world scenarios.

In contrast, PriRAG targets the third setting, which reflects practical deployments where the LLM is cloud-hosted and private knowledge remains local. It addresses the core privacy risk of transmitting retrieved knowledge to the cloud by transforming it into compact proxy knowledge representations before transmission. This representation-level design avoids explicit rewriting and heavy cryptographic operations, enabling effective privacy protection while maintaining efficiency comparable to standard RAG pipelines.

III. METHOD

A. Problem Definition and Threat Model

1) *Problem Definition*: In conventional cloud-based RAG, the LLM is hosted in the cloud, while the private external knowledge source S resides locally on the client. Given a user query q , a retrieval model finds the relevant content e_q from S , which, along with q , is then transmitted to the cloud-based LLM π to generate the final response. While this framework facilitates effective knowledge-enhanced generation, it introduces significant privacy risks due to the direct transmission of e_q in plaintext. Once the retrieved documents e_q leave the local trust boundary and are transmitted to the cloud, sensitive data may be exposed to untrusted entities. In this work, we study the cloud-based RAG framework under privacy constraints, where only a protected alteration of the knowledge can be shared from the local client. This alteration is designed to retain the information essential for generation, while reducing privacy leakage risks even under adversarial attacks.

2) *Attack Surface and Goals*: The attack surface encompasses the protected alteration of retrieved knowledge transmitted to the cloud-based LLM. Based on the adversary capabilities, we categorize the threats into two primary types. (1) *Untargeted Inversion Attack*: The adversary possesses no prior knowledge of specific knowledge instances. However, due to the correlation between the original plaintext knowledge and its protected alteration, the adversary may attempt to reconstruct the knowledge plaintext from the transmitted data. (2) *Targeted Adaptive Attack*: In a more challenging setting, the adversary possesses substantial prior knowledge that links a specific sensitive data field to a particular part of the knowledge representation. Leveraging this knowledge, the attacker adaptively analyzes the transmitted data to infer the exact value of the targeted sensitive field.

B. Methodology Overview

In RAG systems, the prompt usually contains a large volume of retrieved knowledge, typically far exceeding the length of the user query. This redundancy is further exacerbated as more documents are retrieved to enhance knowledge coverage. However, the generated response primarily relies on the model

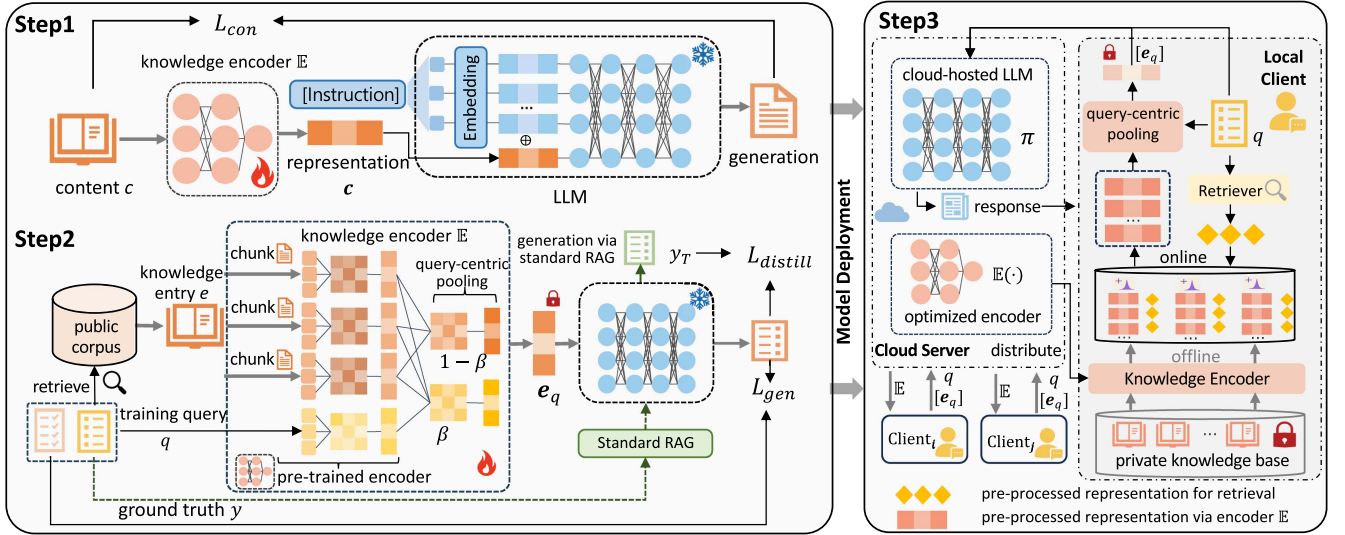


Fig. 1. The framework of PriRAG, which consists of three steps. Step1: representation space alignment for semantic preservation. Step2: query-centric knowledge representation for privacy preservation. Step3: privacy-preserving collaborative inference. Step1 and Step2 correspond to the offline training of the knowledge encoder, while Step3 involves deployment and online inference.

understanding of the query, with the retrieved content serving as supplementary context. Excessive redundant information not only offers limited utility for generation but also enlarges the surface area for potential privacy leakage. Based on these observations, we propose a privacy-preserving framework for cloud-based RAG (Fig. 1) that mitigates privacy leakage while maintaining generation quality for cloud-based RAG systems. Instead of directly sending retrieved documents to the cloud LLM, PriRAG transforms them into compact proxy knowledge representations that preserve task-relevant semantics while obfuscating sensitive details. For example, given a medical record, the proxy knowledge representation retains high-level semantic information but removes identifiable textual content. The LLM then performs inference solely based on these proxy representations, without accessing raw data. This design decouples knowledge utility from raw data exposure, enabling effective generation with reduced privacy risks.

PriRAG centers on compressing retrieved knowledge documents into unified proxy knowledge representations. Its core lies in a knowledge encoder that generates semantic- and privacy-preserving embeddings, aligned with the semantic space of the target LLM to support accurate inference. To train such an encoder, PriRAG comprises two main offline stages. First, we introduce a knowledge construction mechanism that optimizes the alignment between the encoder embedding space and that of the target LLM. Second, to further mitigate potential privacy risks, we propose a query-centric knowledge representation strategy, coupled with a tailored training procedure. Based on these components, we derive a knowledge encoder that can be effectively deployed to enhance the privacy of cloud-RAG systems. Notably, both steps are performed on the cloud server in an offline manner, given the limited computational resources available on local clients. Moreover, the training of the knowledge encoder relies exclusively on publicly available knowledge data, thereby avoiding any additional privacy concerns. Furthermore,

for the model serving phase, we develop a privacy-preserving collaborative inference strategy by incorporating LDP noise into the proxy knowledge representations, thereby providing provable privacy guarantees for all data transmitted during online interactions.

C. Step1: Representation Space Alignment

The proposed knowledge encoder $\mathbb{E}$ is built upon a pre-trained text embedding model to extract semantic information and transform the input into a compact proxy representation. However, a key challenge is that such representation cannot be directly interpreted or utilized by the cloud-based LLM π . To address this, we draw inspiration from recent advances in context compression [17], [36] and introduce a knowledge construction mechanism that bridges the gap between proxy knowledge representations and the LLM. Specifically, given an input content c , the encoder $\mathbb{E}$ produces a proxy embedding $\mathbf{c} = \mathbb{E}(c)$. To render $\mathbf{c}$ interpretable to the LLM, we pair it with a semantically meaningful instruction template t (e.g., "Please provide a reinterpretation of the preceding background text."). This template is encoded using the LLM's tokenizer and embedding layer π' to yield $\mathbf{t}$. The final input to the LLM is the concatenation $\mathbf{c} \oplus \mathbf{t}$. The output of the LLM, denoted y_c , is expected to semantically reconstruct or paraphrase the original input c , thereby validating the informativeness and interpretability of $\mathbf{c}$. Here we define the knowledge construction loss, denoted as $\mathcal{L}_{con}$, which serves as the primary training objective for optimizing the knowledge encoder $\mathbb{E}$.

$$\mathbf{c} = \mathbb{E}(c), \quad \mathcal{L}_{con} = -\sum_{t=1}^T \log P(y_c | c_{<t}, \mathbf{c} \oplus \mathbf{t}; \pi), \quad (1)$$

where $P(\cdot)$ is the conditional probability parameterized by the LLM π , and $\oplus$ denotes the embedding-level concatenation. By minimizing this loss, the encoder is encouraged to generate

proxy knowledge representations that not only retain the semantic fidelity of the original knowledge content but are also aligned with the representation space of the target LLM.

D. Step2: Query-Centric Knowledge Representation

While the knowledge encoder enhanced by the representation space alignment can compress retrieved content to mitigate privacy leakage, it is still inherently agnostic to the target query. Thus, the generated proxy representation may still encode some irrelevant or extraneous information, which not only increases privacy risk but may also degrade response quality. To address this, we develop a query-centric knowledge representation mechanism that selectively emphasizes query-relevant information within the retrieved context e . By dynamically modeling semantic correlation between the query and different segments of the retrieved knowledge, one can reduce the influence of irrelevant content to minimize unnecessary information exposure. Rather than relying on token-level relevance modeling, which is often overly fine-grained and computationally prohibitive, we adopt a hierarchical modeling method. The content e is first segmented into fine-grained chunks $\{a_1^e, \dots, a_m^e\}$, where each chunk a_i^e is a localized semantic unit of the content. We then apply a query-centric attention mechanism to assign a weight w_i to each chunk:

$$w_i = \text{softmax}(f(\mathbf{a}_i^e, \mathbf{q})), \quad \mathbf{e}_q = \sum_{i=1}^m w_i \cdot \mathbf{a}_i^e, \quad (2)$$

where $\sum_{i=1}^m w_i = 1$ and $\mathbf{a}_i^e = \mathbb{E}(a_i^e)$ denotes the proxy representation of chunk a_i , and $f(\cdot, \cdot)$ is a similarity function between the chunk and the query. The softmax function normalizes attention weights across all chunks. This selective aggregation produces a final proxy representation $\mathbf{e}_q$ that retains the most semantically relevant information for the given query, reducing both privacy leakage and generation noise.

To further ensure that the LLM can effectively leverage the compressed proxy representation $\mathbf{e}_q$ for downstream tasks, we introduce a privacy-preserving training strategy for optimizing knowledge encoder $\mathbb{E}$. First, we fine-tune the encoder $\mathbb{E}$ such that, when provided with the original query q and its corresponding compressed proxy knowledge representation $\mathbf{e}_q$, the LLM generates responses that are semantically aligned with a reference answer y . This procedure explicitly transforms sensitive knowledge into proxy knowledge representations interpretable by the LLM, thereby preserving generation quality while preventing exposure of raw text. Formally, we introduce $\mathcal{L}_{\text{gen}}$ that quantifies the alignment with answer y :

$$\begin{aligned} \mathbf{e}_q &= \mathbb{E}(q, e), \quad \mathbf{q} = \pi'(q), \\ \mathcal{L}_{\text{gen}} &= -\sum_{t=1}^T \log P(y_t | y_{<t}, \mathbf{q} \oplus \mathbf{e}_q; \pi), \end{aligned} \quad (3)$$

where the operator $\oplus$ denotes the concatenation of the compressed proxy representation $\mathbf{e}_q$ with the query embedding sequence $\mathbf{q}$ at the token embedding layer of the LLM.

Second, we adopt a knowledge distillation approach in which a full RAG pipeline (i.e., using uncompressed retrieved

knowledge) serves as a teacher model. The knowledge encoder is trained to allow LLMs to mimic the outputs y_T of the teacher while operating on privacy-preserving proxy representation. This distillation process encourages the encoder to focus on query-relevant semantics, promoting robust generation performance while mitigating the exposure of sensitive information. We formalize this objective $\mathcal{L}_{\text{distill}}$ by minimizing the divergence between the output distributions of the LLM conditioned on raw retrieved knowledge e and those conditioned on the proxy representation $\mathbf{e}_q$:

$$\begin{aligned} P_t &= P(y_t | y_{<t}, q + e; \pi), \quad \hat{P}_t = P(y_t | y_{<t}, \mathbf{q} \oplus \mathbf{e}_q; \pi), \\ \mathcal{L}_{\text{distill}} &= D_{\text{KL}} \left(\sum_{t=1}^T \log P_t \parallel \sum_{t=1}^T \log \hat{P}_t \right), \end{aligned} \quad (4)$$

where $D_{\text{KL}}(\cdot \parallel \cdot)$ denotes the Kullback–Leibler divergence and $+$ denotes the token-level concatenation. The overall training objective $\mathcal{L}_f$ combines both generation and distillation losses with λ to balance them: $\mathcal{L}_f = \mathcal{L}_{\text{gen}} + \lambda \cdot \mathcal{L}_{\text{distill}}$.

Recognizing that $\mathbb{E}$ may struggle to capture all task-relevant semantics for the query through strict query-centric compression, we introduce a hybrid training strategy. During training, we construct a mixed input consisting of a weighted combination of the query-centric proxy representation and a standard proxy representation based on hyper-parameter β . This mixture allows the encoder to retain general semantic context while prioritizing query-specific relevance, ultimately enhancing the LLM's ability to produce accurate and contextually grounded responses. Moreover, it is worth noting that the knowledge encoder can be effectively trained by public knowledge bases and query datasets, even when their distributions differ from those of the private knowledge and query tasks. This design mitigates privacy risks during the training phase.

E. Step3: Privacy Preserving Collaborative Inference

Next, we present our proposed privacy-preserving collaborative online inference strategy (Step 3 in Fig. 1). Following the offline optimization of the knowledge encoder described earlier, the optimized knowledge encoder is deployed to the client side for subsequent inference. For each knowledge entry e , the client employs the encoder to compute fine-grained representations $\{\mathbf{a}_i^e\}_{i=1}^m$, for its constituent chunks, $\{a_i^e\}_{i=1}^m$. As these representations may encode sensitive information, transmitting them directly to the server risks privacy leakage. To mitigate this, we introduce a perturbed version of the representations by injecting Gaussian noise:

$$\hat{\mathbf{a}}_i^e = \text{clip}(\mathbf{a}_i^e, \Delta) + \mathbf{n}_i, \quad \mathbf{n}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d), \quad \mathbf{a}_i^e \in \mathbb{R}^d. \quad (5)$$

Here, $\text{clip}(\cdot)$ constrains the ℓ_2 -norm of $\mathbf{a}_i^e$ within a threshold Δ , and $\mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$ denotes a d -dimensional Gaussian distribution with zero mean and isotropic covariance, where σ^2 is the noise variance. This perturbation mechanism [37], [38] provides formal privacy guarantees for the released representations $\{\hat{\mathbf{a}}_i^e\}_{i=1}^m$, thereby protecting any sensitive information encoded in the original plaintext of a_i . Notably, this encoding and perturbation process is performed only once in an offline manner, thereby

Algorithm 1: Client-Cloud Collaborative Inference Workflow.

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[CLLOUD]: Distribute optimized encoder  $\mathbb{E}(\cdot)$  to client.
// Offline Data Pre-processing
[CLIENT]: Deploy the knowledge encoder  $\mathbb{E}(\cdot)$ .
for all  $e \in D$  do
    [CLIENT]: Split  $e$  into fine-grained chunks  $\{a_i^e\}_{i=1}^m$ .
    [CLIENT]: Compute representation  $\mathbf{a}_i^e = \mathbb{E}(a_i^e)$ .
    [CLIENT]: Perturb  $\mathbf{a}_i^e$  to  $\hat{\mathbf{a}}_i^e$  via (5).
end for
// Online Collaborative Inference
[CLIENT]: Retrieve knowledge entry  $e$  based on query  $q$ .
[CLIENT]: Derive query-centric  $[\mathbf{e}_q]$  via (6).
[CLIENT]: Send representation  $[\mathbf{e}_q]$  and  $q$  to the cloud.
[CLLOUD]: Utilize LLM to generate response  $y$ .
[CLLOUD]: Send  $y$  to client.

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minimizing the computational cost of subsequent online interactions between the client and the server.

After constructing private representations for all knowledge entries, PriRAG enables privacy-preserving retrieval-augmented generation services. Specifically, given a query q , the system first retrieves the most relevant knowledge entries e from the knowledge base using any standard text retrieval method [39]. Then, following (2), a lightweight attention module within the knowledge encoder aggregates the perturbed chunk representations:

$$[\mathbf{e}_q] = \sum_{i=1}^m w_i \hat{\mathbf{a}}_i^e = \sum_{i=1}^m w_i \mathbf{a}_i^e + \sum_{i=1}^m w_i \mathbf{n}_i, \quad (6)$$

where $w_i = \text{softmax}(f(\hat{\mathbf{a}}_i^e, \mathbf{q}))$ and $\mathbf{e}_q$ denotes the privacy-preserving representation of entry e conditioned on the query q , $\mathbf{q}$ is the query embedding, and $f(\cdot, \cdot)$ is a similarity function. Finally, the private representation $[\mathbf{e}_q]$ and the query q are transmitted to the cloud server, which concatenates $[\mathbf{e}_q]$ with the token embeddings of $\mathbf{q}$ and performs inference via the cloud-based LLM (Algorithm 1).

Next, we present the formal privacy guarantee formulation of our method. In mainstream RAG approaches [39], the raw data are typically partitioned into multiple knowledge chunks, which are subsequently utilized as the basic unit for retrieval and prompt construction. Accordingly, we analyze the privacy guarantee for each individual knowledge chunk a through the lens of differential privacy. Specifically, in PriRAG, any chunk a undergoes a processing pipeline: $a \rightarrow \mathbf{a} \rightarrow \hat{\mathbf{a}} \rightarrow [\mathbf{e}_q]$, where only the final proxy representation $[\mathbf{e}_q]$ is transmitted to the cloud for further processing. Based on the data processing inequality [40], the representation $\hat{\mathbf{a}}$ cannot reveal more private information than the original chunk a . Therefore, the differential privacy guarantee established for $\hat{\mathbf{a}}$ also extends to the original chunk a , as formally stated below.

Lemma 1: Let a be any knowledge chunk which is the basic data unit used in the RAG system, and $\hat{\mathbf{a}}$ be its perturbed representation via (5), with noise variance

$$\sigma^2 = \frac{8 \ln(1.25/\delta) \cdot \Delta^2}{\epsilon^2}. \quad (7)$$

Then both the mechanisms $a \mapsto \hat{\mathbf{a}}$ and $a \mapsto [\mathbf{e}_q]$ through $a \rightarrow \hat{\mathbf{a}} \rightarrow [\mathbf{e}_q]$ satisfy (ϵ, δ) -LDP for the original chunk a .

Moreover, in practical RAG deployments, the same knowledge chunk may be repeatedly retrieved by adversaries issuing diverse queries, potentially leading to cumulative privacy budget consumption and, consequently, weakened privacy guarantees. To address this issue, the PriRAG method adopts a *shared-noise mechanism* to perturb the representation of each knowledge chunk only once. Specifically, the client computes a single perturbed version of the representation for each knowledge chunk, which is then reused across all queries referencing that data. This mechanism ensures that PriRAG preserves the original (ϵ, δ) -LDP guarantee, even under repeated query access. Formally, let q_i denote any query that adversaries utilize to retrieve the target knowledge data, and let set $\mathcal{Q} = \{q_i\}_{i=1}^k$ represent all such queries, where k is the total number of queries. We establish the following lemma.

Lemma 2: Assume that an adversary obtains a set of representations $\{[\mathbf{e}_q] \mid q \in \mathcal{Q}\}$ associated with the same knowledge data for the purpose of a privacy attack. Under the shared-noise mechanism employed by PriRAG, the original (ϵ, δ) -LDP guarantee for the chunk a remains valid.

Besides, we emphasize that it is highly efficient to apply the LDP technique in our PriRAG method. In a naive LDP-enhanced RAG setting, the client can directly leverage the LLM embedding layer to encode the knowledge chunk a , and then privatize the token-level embedding sequence $\mathbf{W}_a \in \mathbb{R}^{l \times d}$, where l is the number of tokens in the chunk a . This suffers from the curse of dimensionality inherent to LDP, as the large number of elements to be privatized necessitates injecting substantial noise, which severely compromises utility. In contrast, PriRAG compresses each knowledge chunk into a unified proxy representation $\mathbf{a} \in \mathbb{R}^{1 \times d}$, thereby significantly reducing the dimensionality of the data to be privatized. Under the same privacy budget, this compression substantially lowers the noise introduced to each vector element, thus improving utility. We formally quantify this advantage in Lemma 3.

Lemma 3: To satisfy the same (ϵ, δ) -LDP guarantee, the required noise variances σ_W^2 and σ_a^2 for perturbing $\mathbf{W}_a$ and $\mathbf{a}$, respectively, satisfy: $\sigma_a^2 = \frac{1}{l} \sigma_W^2$.

This analysis shows that our method provides more efficient privacy protection than conventional token-level LDP methods, especially for knowledge entries with a large amount of content. Therefore, the proposed proxy knowledge representation framework effectively enhances the privacy-utility trade-off, enabling strong privacy guarantees while maintaining utility for long knowledge entries. These properties make the approach particularly well-suited for practical deployment of LDP mechanisms in collaborative inference scenarios. The detailed proofs of Lemmas 1, 2, and 3 are provided in Appendix B, available online.

F. Discussion on the Practicality of PriRAG

PriRAG can be seamlessly deployed in real-world RAG systems that require privacy-sensitive knowledge sharing with cloud-based LLMs and demonstrates strong practicality for deployment in real-world settings. On the commercial side, LLM

providers can integrate our framework into closed-source platforms to address increasing user concerns regarding data privacy in RAG services [11]. Such integration enables service providers to enhance the reliability and trustworthiness of their offerings without requiring access to raw user data. Consequently, providers can attract a broader user base, particularly those who might otherwise be hesitant to adopt cloud-based LLM services due to data security risks. Meanwhile, the rapid development of increasingly capable open-source LLMs creates opportunities for the PriRAG framework to further promote the construction of privacy-preserving RAG models. Besides, we have already released implementations of PriRAG for Qwen series LLMs. This not only enriches the open-source ecosystem of privacy-preserving reasoning models but also empowers individual users and organizations with limited local computational resources to leverage PriRAG in a secure, cloud-based setting. This dual applicability underscores the versatility of our approach across enterprise-grade and community-driven ecosystems.

From a technical perspective, PriRAG provides tangible efficiency and usability benefits in addition to privacy protection. The knowledge encoder can be trained solely on publicly available corpora, avoiding any risk of private data leakage during model development. At inference time, proxy knowledge representations compress lengthy retrieved documents into compact vectors, which reduces latency and improves responsiveness compared to standard RAG pipelines. Moreover, the encoder adaptation to a target LLM requires fine-tuning only a small set of parameters (e.g., 20–50 MB) using tens of thousands of documents. This lightweight cost is negligible relative to fine-tuning a full LLM, making PriRAG feasible even under constrained computational budgets. A practical limitation is that most commercial LLM providers only expose text-based interfaces, which constrains immediate large-scale deployment. Nevertheless, PriRAG remains readily applicable in controlled or enterprise-hosted environments and highlights a promising pathway toward more flexible API designs for privacy-sensitive applications.

IV. EXPERIMENT

A. Experimental Settings

1) *Datasets and Models*: During the knowledge construction phase, we utilized the Wikipedia corpus up to December 2021. We randomly selected 200 k entries from 33.1 million records for pretraining. For the query-centric knowledge representation training phase, we employed 37 k question-answer pairs from the PwC and SQuAD v2 datasets. These corpora help ensure that the proxy knowledge representation can be effectively leveraged by LLMs to enhance generation quality in downstream tasks. In the test phase, we conducted experiments on four classes of benchmarks: (1) Open-Domain: TriviaQA [41] (2) Complex Reasoning: HotpotQA [42], 2WikiMultiHopQA (2WikiQA) [43], StrategyQA [44] (3) Professional field on Medical: PubMedQA [45] and BioASQ [46] (4) Factual Integrity Evaluation: TruthfulQA [47]. These benchmarks cover a wide spectrum of QA tasks, from general open-domain retrieval to multi-hop reasoning, medical knowledge, and factual consistency,

allowing us to comprehensively evaluate the effectiveness and generalization of PriRAG under diverse and privacy-sensitive scenarios. It is noted that the evaluation datasets are disjoint from those training datasets.

To evaluate the generalization of the proposed PriRAG, we utilize multiple model scales and architectures, including Mistral-7B-Instruct-v0.2 [48], Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct and Qwen2.5-32B-Instruct [49]. The encoder is initialized by the embedding model SFR-Embedding-Mistral [50]. We adopt the Adam optimizer for all training phases, using consistent configurations with a batch size of 4, a warmup ratio of 0.03, a hyper-parameter β of 0.1 and one training epoch. The learning rate is set to 6.0×10^{-3} for the knowledge construction task and 2.0×10^{-5} for the query-centric training phase. The hyperparameter λ , which balances the generation and distillation losses, is fixed at 2.

2) *Baseline Methods*: We conducted a series of experiments using several representative baseline methods:

- *Standard pipeline*: w/o RAG directly utilizes the LLMs to generate responses without any retrieval knowledge. RAG [12] follows the standard paradigm in which the client directly transmits the plaintext of retrieved knowledge to cloud-based LLMs. Both methods serve as a basic reference for assessing privacy-utility trade-offs.
- *Rewriting-based method*: SAGE [13] is implemented as a generative transformation pipeline that replaces the original retrieved context with a rewritten surrogate. For each retrieved document, we first use an LLM-based extractor (e.g., DeepSeek-V3.2) to extract salient attribute-level information, preserving core semantics while removing surface details. These attributes are then fed into an LLM to regenerate a coherent context, which is used as the input to the downstream model.
- *Sanitization-based method*: Methods such as those proposed in [22], [23] employ synonym substitution and lexical transformations to obscure sensitive terms in the retrieved information, thereby enhancing privacy while attempting to retain overall utility.
- *Differential privacy-based method*: We implement a LDP-RAG baseline which injects Gaussian noise into token-level embeddings of retrieved knowledge following [16], ensuring (ϵ, δ) -LDP before input to the LLM. Since the global sensitivity of embeddings is not directly tractable, we follow standard practice [37], [51] by enforcing an ℓ_2 norm bound (e.g., unit clipping) and setting the sensitivity to 1.0, with δ fixed to 10^{-5} . Noise in PriRAG is calibrated based on (7) under the same privacy budget ϵ , enabling fair comparison. Under a fixed noise level, LDP-RAG generally requires substantially larger ϵ values to maintain acceptable utility, as implied by Lemma 3 and shown in Fig. 4.

B. Performance Evaluation

We conduct a comparative analysis of PriRAG against several baseline methods across multiple benchmarks (Table I). The results demonstrate the practical feasibility of using proxy

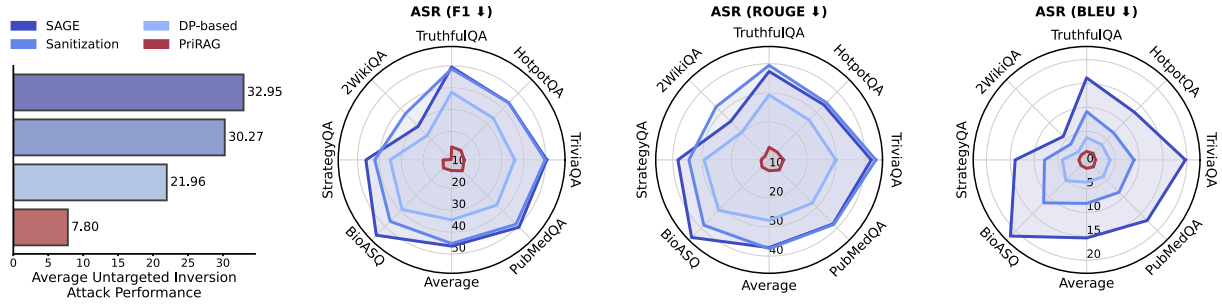


Fig. 2. The attack success rate (ASR) of different methods under untargeted inversion attack.

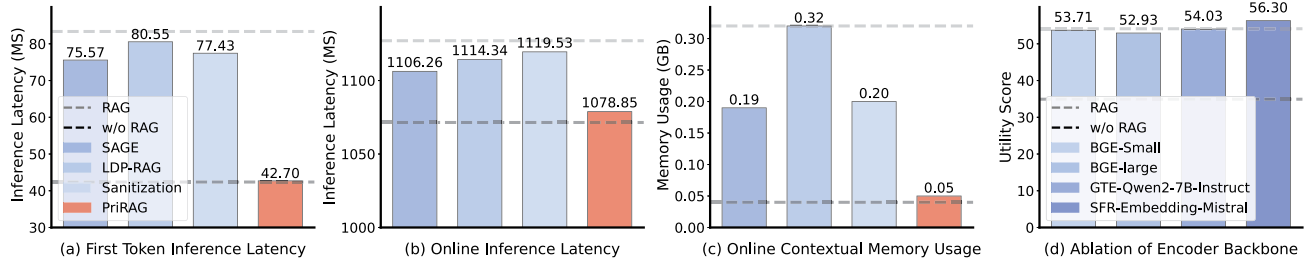


Fig. 3. Analysis of PriRAG: (a) first-token inference latency, (b) online inference latency, (c) contextual memory usage during online inference, and (d) generalization of the knowledge encoder.

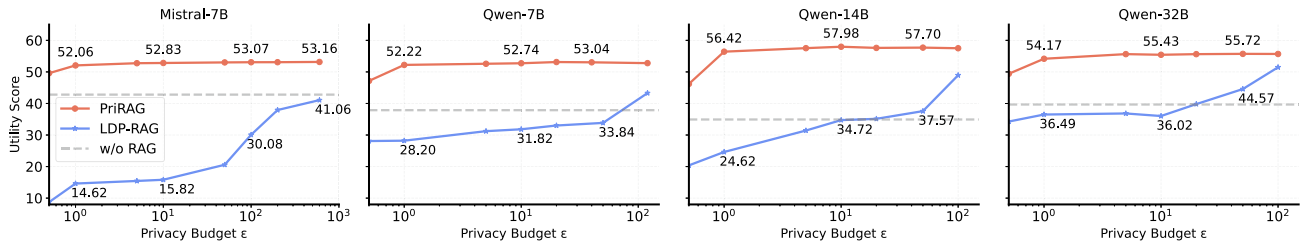


Fig. 4. Performance comparison of various methods under the same privacy budget (ϵ) across different models.

knowledge representations as a medium for sharing knowledge between private knowledge owners and cloud-based LLMs.

First, the experimental results clearly demonstrate PriRAG's strong capability in knowledge preservation and utility enhancement. For example, PriRAG significantly outperforms the w/o RAG baseline, with improvements of up to 21.11, confirming that the proxy representations effectively capture and retain essential query-relevant information. Moreover, PriRAG matches or even surpasses traditional RAG pipelines on several benchmarks. This suggests that proxy embeddings not only preserve critical semantics but also filter out irrelevant or redundant context through the query-centric mechanism, thereby reducing noise and improving both inference performance and generation quality. Performance gains are more pronounced in tasks where answers depend on a focused subset of evidence, while benchmarks involving multi-hop reasoning or higher redundancy may exhibit slight fluctuations due to compression. Nevertheless, PriRAG maintains consistently strong performance across all

tasks. Second, PriRAG effectively mitigates the performance degradation associated with privacy protection. For example, compared with the DP-based methods, PriRAG maintains high generation quality even under a stricter privacy budget ($\epsilon = 1$), while the LDP-RAG with a much weaker privacy guarantee ($\epsilon = 80$) suffers greater performance degradation. This highlights the advantage of compressing raw knowledge into structured proxy knowledge representations, which significantly reduce sensitivity to noise and allow for calibrated DP perturbation with minimal utility loss. Compared with Non-DP baselines SAGE and Sanitization, PriRAG shares only filtered and compressed proxy information. This inherent design fundamentally reduces the leakage of sensitive content, offering a far more robust and formally guaranteed privacy protection. Third, PriRAG's architectural design endows it with exceptional model-agnosticism and broad generalization. This flexibility empowers users to choose their preferred backend LLM without necessitating a redesign or re-training of the privacy protection mechanism.

TABLE I

PERFORMANCE EVALUATION OF DIFFERENT METHODS. U_{avg} DENOTES THE AVERAGED UTILITY SCORE. *PLAIN* INDICATES WHETHER PART OF UNPROTECTED CONTENT IS SHARED IN PLAINTEXT, WHICH IS HIGHLY PRIVACY SENSITIVE. (THE CHOICE OF ϵ VALUE IS SHOWN IN THE APPENDIX.).

LLM	Method	TriviaQA	HotpotQA	TruthfulQA	2WikiQA	StrategyQA	BioASQ	PubMedQA	$U_{avg} \uparrow$	<i>Plain</i>
Mistral-7B	w/o RAG	52.30 $\pm$ 0.37	24.20 $\pm$ 0.33	35.25 $\pm$ 0.25	34.47 $\pm$ 0.80	50.50 $\pm$ 0.14	35.96 $\pm$ 1.06	61.70 $\pm$ 1.28	42.05	×
	RAG	64.20 $\pm$ 0.36	39.73 $\pm$ 0.21	33.55 $\pm$ 0.87	69.13 $\pm$ 0.31	73.27 $\pm$ 0.52	70.54 $\pm$ 0.71	89.00 $\pm$ 0.59	62.78	✓
	SAGE	62.43 $\pm$ 0.26	36.87 $\pm$ 0.90	33.60 $\pm$ 0.23	63.17 $\pm$ 0.66	74.97 $\pm$ 0.29	67.79 $\pm$ 2.14	91.57 $\pm$ 0.48	61.48	✓
	Sanitization	51.93 $\pm$ 0.95	26.90 $\pm$ 0.62	32.04 $\pm$ 0.46	40.30 $\pm$ 1.02	61.23 $\pm$ 0.61	52.56 $\pm$ 0.63	74.83 $\pm$ 1.53	48.54	✓
	LDP-RAG $\epsilon=700$	55.77 $\pm$ 0.54	28.10 $\pm$ 0.36	34.30 $\pm$ 0.31	34.93 $\pm$ 0.09	58.70 $\pm$ 0.85	42.32 $\pm$ 1.86	61.40 $\pm$ 2.62	45.08	×
	PriRAG $\epsilon=1$	59.00 $\pm$ 0.54	26.67 $\pm$ 0.26	33.27 $\pm$ 0.24	34.63 $\pm$ 0.41	69.13 $\pm$ 1.94	56.06 $\pm$ 0.17	87.57 $\pm$ 1.27	52.33	×
Qwen-7B	w/o RAG	48.37 $\pm$ 0.25	23.80 $\pm$ 0.22	33.48 $\pm$ 0.43	32.80 $\pm$ 0.36	59.23 $\pm$ 0.25	31.83 $\pm$ 2.01	35.57 $\pm$ 0.92	37.87	×
	RAG	59.60 $\pm$ 0.29	34.70 $\pm$ 0.36	33.41 $\pm$ 0.15	49.93 $\pm$ 0.45	80.30 $\pm$ 0.08	63.42 $\pm$ 0.88	74.10 $\pm$ 0.57	56.49	✓
	SAGE	58.50 $\pm$ 0.43	33.13 $\pm$ 0.46	32.97 $\pm$ 0.45	48.53 $\pm$ 0.33	79.70 $\pm$ 0.14	61.30 $\pm$ 0.17	73.50 $\pm$ 1.27	55.38	✓
	Sanitization	48.13 $\pm$ 0.29	24.93 $\pm$ 0.21	29.72 $\pm$ 0.26	41.70 $\pm$ 0.94	68.10 $\pm$ 0.78	50.19 $\pm$ 1.86	53.97 $\pm$ 0.92	45.25	✓
	LDP-RAG $\epsilon=120$	47.33 $\pm$ 1.06	24.97 $\pm$ 0.87	34.44 $\pm$ 0.40	41.63 $\pm$ 0.85	67.93 $\pm$ 0.26	46.69 $\pm$ 2.78	40.03 $\pm$ 1.03	43.29	×
	PriRAG $\epsilon=1$	52.03 $\pm$ 1.18	25.50 $\pm$ 0.22	37.95 $\pm$ 0.31	44.70 $\pm$ 0.70	70.00 $\pm$ 0.85	50.56 $\pm$ 0.53	84.77 $\pm$ 1.78	52.22	×
Qwen-14B	w/o RAG	58.93 $\pm$ 0.21	27.40 $\pm$ 0.37	32.74 $\pm$ 0.50	38.07 $\pm$ 0.62	35.07 $\pm$ 1.19	37.05 $\pm$ 1.44	16.47 $\pm$ 1.30	35.10	×
	RAG	61.80 $\pm$ 0.22	39.27 $\pm$ 0.45	29.81 $\pm$ 0.31	72.67 $\pm$ 0.54	61.63 $\pm$ 0.45	63.54 $\pm$ 1.27	48.93 $\pm$ 0.54	53.95	✓
	SAGE	60.63 $\pm$ 0.33	36.37 $\pm$ 0.19	27.16 $\pm$ 0.62	59.90 $\pm$ 0.22	49.87 $\pm$ 0.92	57.43 $\pm$ 1.16	56.30 $\pm$ 0.24	49.67	✓
	Sanitization	53.87 $\pm$ 0.96	29.17 $\pm$ 0.73	27.16 $\pm$ 0.83	39.90 $\pm$ 0.64	59.80 $\pm$ 0.49	46.82 $\pm$ 1.10	19.13 $\pm$ 0.25	39.41	✓
	LDP-RAG $\epsilon=80$	55.10 $\pm$ 0.62	24.40 $\pm$ 0.45	34.18 $\pm$ 0.23	36.50 $\pm$ 0.37	54.93 $\pm$ 0.69	41.20 $\pm$ 0.92	15.90 $\pm$ 1.14	37.46	×
	PriRAG $\epsilon=1$	59.83 $\pm$ 0.33	30.10 $\pm$ 0.43	38.95 $\pm$ 0.30	47.63 $\pm$ 0.45	75.27 $\pm$ 0.65	57.43 $\pm$ 1.23	84.23 $\pm$ 0.33	56.21	×
Qwen-32B	w/o RAG	55.57 $\pm$ 0.29	25.53 $\pm$ 0.62	34.03 $\pm$ 0.49	39.90 $\pm$ 0.71	70.73 $\pm$ 0.17	30.34 $\pm$ 1.91	24.27 $\pm$ 0.26	40.05	×
	RAG	62.37 $\pm$ 0.09	38.70 $\pm$ 0.22	19.59 $\pm$ 0.22	70.50 $\pm$ 0.41	87.23 $\pm$ 0.19	61.05 $\pm$ 0.31	75.57 $\pm$ 0.26	59.29	✓
	SAGE	62.77 $\pm$ 0.12	37.53 $\pm$ 0.57	22.97 $\pm$ 0.45	60.03 $\pm$ 0.39	84.13 $\pm$ 0.19	62.92 $\pm$ 1.06	79.07 $\pm$ 0.29	58.49	✓
	Sanitization	54.57 $\pm$ 0.45	28.10 $\pm$ 0.33	12.48 $\pm$ 0.58	48.73 $\pm$ 1.30	72.87 $\pm$ 0.62	44.19 $\pm$ 1.70	37.20 $\pm$ 0.29	42.59	✓
	LDP-RAG $\epsilon=80$	54.77 $\pm$ 0.83	24.40 $\pm$ 0.14	27.36 $\pm$ 0.77	40.80 $\pm$ 0.93	71.00 $\pm$ 0.36	46.19 $\pm$ 1.68	47.87 $\pm$ 0.26	44.63	×
	PriRAG $\epsilon=1$	59.10 $\pm$ 0.62	28.90 $\pm$ 0.73	38.99 $\pm$ 1.17	46.30 $\pm$ 0.28	70.80 $\pm$ 1.14	54.18 $\pm$ 1.74	79.50 $\pm$ 3.68	53.97	×

Furthermore, PriRAG exhibits robust compatibility with various retrieval data sources. These attributes collectively highlight PriRAG’s versatility and scalability for broad deployment in real-world, cloud-native RAG applications, offering a reliable privacy-preserving solution across diverse industries and use cases.

C. Empirical Privacy Evaluation

In addition to the theoretical privacy guarantees discussed in Section III-E, we empirically assess potential privacy leakage risks under an explicit threat model. We instantiate this threat model in concrete attack settings. Specifically, we assume an honest-but-curious adversary who observes the query and the transmitted proxy representations, and interacts with the system through black-box access to the LLM. The adversary does not have access to the original knowledge or the internal parameters of the knowledge encoder. The goal of the adversary is to infer sensitive information contained in the retrieved documents beyond what is required for answering the query.

First, in the untargeted inversion attack setting, we assume that the adversary possesses no prior knowledge of the retrieved original content and attempts to reconstruct the original textual knowledge solely from the shared proxy information transmitted. To simulate this scenario, we designed specific inversion-style instructions (detailed in Prompt 1 in the Appendix C, available online) and directly applied them to the LLM, using the proxy representation generated by the optimized knowledge encoder within PriRAG. The generated outputs were then compared against the original textual knowledge, with the degree of sensitive content recovery quantified using established

text similarity metrics, including F1 score, ROUGE, and BLEU. The experimental results in Fig. 2 clearly demonstrate that PriRAG exhibits strong resistance to untargeted inversion attacks. Compared to all baseline methods, significantly less original plaintext could be recovered from PriRAG’s outputs. These findings indicate that the proxy knowledge representations not only preserve downstream task performance but also provide robust protection against inversion-based privacy threats.

Second, to simulate a more advanced and realistic attack scenario, we further considered targeted adaptive attacks. In this setting, the adversary possesses auxiliary knowledge and can perform targeted adaptive attacks. We implemented such attacks by crafting auxiliary questions, derived from the retrieved context using GPT-4o (see Prompt 2 and a representative case study is provided in Appendix C, available online). These auxiliary questions are semantically related to the retrieved documents but differ from the original user queries, with the explicit aim of probing for sensitive information inadvertently embedded within the proxy knowledge representations. The goal is to evaluate whether the proxy representation inadvertently discloses information beyond what is required to answer the original query. As shown in Table II, PriRAG achieves a more favorable utility–privacy trade-off than baseline methods. For instance, while the LDP-RAG method offers strong privacy, it suffers from substantial degradation in generation performance, thus limiting its practical utility. Conversely, the SAGE method preserves performance closer to standard RAG but exhibits substantially higher information leakage, as it transmits partially redacted plaintext. PriRAG’s query-centric and compressed proxy knowledge representations effectively limit unnecessary knowledge exposure while preserving task performance. This design enables it to

TABLE II

PERFORMANCE OF DIFFERENT METHODS UNDER TARGETED ADAPTIVE ATTACKS ACROSS EACH BENCHMARK. P_{avg} DENOTES THE AVERAGE ATTACK SUCCESS RATE, EVALUATED USING MODEL PERFORMANCE ON ADAPTIVELY CRAFTED QUERIES. TRADE-OFF BETWEEN UTILITY AND PRIVACY IS MEASURED AS $U_{avg} - P_{avg}$, WHERE U_{avg} REPRESENTS THE AVERAGED UTILITY SCORE REPORTED IN TABLE I.

LLM	Method	TriviaQA	HotpotQA	TruthfulQA	2WikiQA	StrategyQA	BioASQ	PubMedQA	$P_{avg} \downarrow$	Trade-off $\uparrow$
Mistral-7B	SAGE	60.23 $\pm$ 0.69	62.79 $\pm$ 0.83	18.95 $\pm$ 4.55	61.14 $\pm$ 0.14	12.70 $\pm$ 1.08	21.43 $\pm$ 0.76	10.81 $\pm$ 0.51	35.44	26.04
	Sanitization	20.86 $\pm$ 1.37	22.43 $\pm$ 5.21	14.74 $\pm$ 3.10	19.63 $\pm$ 2.67	4.97 $\pm$ 1.90	4.42 $\pm$ 0.45	2.70 $\pm$ 2.52	12.82	35.72
	LDP-RAG	18.89 $\pm$ 3.44	22.08 $\pm$ 2.76	3.51 $\pm$ 2.16	16.17 $\pm$ 3.29	3.08 $\pm$ 0.55	2.71 $\pm$ 1.25	0.68 $\pm$ 0.96	9.59	35.49
	PriRAG	13.77 $\pm$ 2.16	11.21 $\pm$ 3.49	10.88 $\pm$ 2.16	11.78 $\pm$ 1.77	3.45 $\pm$ 0.90	2.72 $\pm$ 0.50	3.71 $\pm$ 0.46	8.22	44.11
Qwen-7B	SAGE	59.14 $\pm$ 0.72	62.93 $\pm$ 0.92	14.04 $\pm$ 0.50	60.86 $\pm$ 0.54	11.53 $\pm$ 0.78	19.97 $\pm$ 0.95	9.18 $\pm$ 0.42	33.95	21.43
	Sanitization	30.75 $\pm$ 0.99	18.01 $\pm$ 2.83	10.18 $\pm$ 1.98	19.22 $\pm$ 2.96	5.77 $\pm$ 0.08	6.64 $\pm$ 1.78	5.10 $\pm$ 0.81	13.67	31.58
	LDP-RAG	22.56 $\pm$ 1.32	22.06 $\pm$ 0.90	11.58 $\pm$ 4.79	17.13 $\pm$ 1.92	5.77 $\pm$ 0.08	5.95 $\pm$ 1.01	5.44 $\pm$ 0.44	12.93	30.36
	PriRAG	8.17 $\pm$ 2.51	4.77 $\pm$ 1.36	10.18 $\pm$ 2.16	8.80 $\pm$ 1.11	3.44 $\pm$ 1.87	1.05 $\pm$ 0.01	3.06 $\pm$ 0.85	5.64	46.58
Qwen-14B	SAGE	53.84 $\pm$ 3.00	61.61 $\pm$ 0.73	10.88 $\pm$ 4.73	52.28 $\pm$ 2.76	15.70 $\pm$ 0.95	18.88 $\pm$ 0.70	8.16 $\pm$ 0.83	31.62	18.05
	Sanitization	23.08 $\pm$ 1.79	19.59 $\pm$ 1.30	9.82 $\pm$ 6.22	21.13 $\pm$ 5.20	6.89 $\pm$ 0.90	7.33 $\pm$ 2.22	3.40 $\pm$ 0.96	13.04	26.37
	LDP-RAG	9.26 $\pm$ 2.44	13.29 $\pm$ 2.03	5.96 $\pm$ 2.16	14.74 $\pm$ 1.65	5.77 $\pm$ 0.95	5.56 $\pm$ 0.49	4.42 $\pm$ 0.48	8.43	29.03
	PriRAG	10.93 $\pm$ 1.00	9.55 $\pm$ 1.69	7.02 $\pm$ 3.47	7.80 $\pm$ 0.67	6.12 $\pm$ 1.04	1.75 $\pm$ 0.50	2.38 $\pm$ 0.48	6.51	49.70
Qwen-32B	SAGE	56.01 $\pm$ 2.24	64.14 $\pm$ 1.95	12.63 $\pm$ 2.27	60.87 $\pm$ 1.18	12.50 $\pm$ 0.62	18.75 $\pm$ 0.85	9.70 $\pm$ 0.43	33.51	24.98
	Sanitization	29.15 $\pm$ 4.90	24.37 $\pm$ 2.52	7.72 $\pm$ 2.16	19.81 $\pm$ 1.81	5.85 $\pm$ 1.64	7.64 $\pm$ 0.98	8.03 $\pm$ 0.82	14.65	27.94
	LDP-RAG	10.40 $\pm$ 2.21	13.35 $\pm$ 1.94	5.61 $\pm$ 1.98	13.53 $\pm$ 4.16	3.90 $\pm$ 1.10	4.17 $\pm$ 0.00	5.35 $\pm$ 0.46	8.04	36.59
	PriRAG	5.77 $\pm$ 2.82	8.21 $\pm$ 3.14	6.67 $\pm$ 2.16	3.38 $\pm$ 0.68	3.90 $\pm$ 1.44	2.43 $\pm$ 0.98	4.01 $\pm$ 0.02	4.91	49.06

effectively withstand targeted adaptive attacks by presenting the LLM with only the most relevant, privacy-preserving, and minimized set of information pertinent to the current query. Consequently, PriRAG is proven to be a practical and effective solution for constructing privacy-compliant RAG systems.

D. Efficiency Analysis

Beyond its privacy guarantees and enhanced utility, the practical viability of PriRAG is further underscored by its efficiency, both during online inference and offline training.

To demonstrate the practicality of PriRAG, we first conducted a thorough comparison of first-token inference latency, online inference time (Fig. 3), and contextual memory usage during online inference across various methods. PriRAG achieves superior inference efficiency compared to all competing methods. This significant advantage stems directly from its core mechanism. By compressing retrieved content into compact proxy knowledge representations, PriRAG reduces the input token length presented to the LLM. This not only minimizes the computational load on the LLM during generation but also results in online latency comparable to, or even matching, the highly efficient w/o RAG baseline.

Second, the PriRAG framework necessitates the optimization of the knowledge encoder to align with the dual objectives of privacy preservation and integration with the target LLM. Importantly, this optimization process is both efficient and resource-friendly. Our approach fine-tunes only a small subset of parameters within the knowledge encoder, while keeping the underlying, typically much larger, LLM architecture frozen. The number of trainable parameters involved in this process is notably minimal, generally ranging from 20 MB to 50 MB, depending on the specific LLM architecture and the chosen encoder configuration. Furthermore, this fine-tuning uses a relatively small corpus (tens of thousands of documents), far smaller than typical LLM pre-training datasets. This modest data

TABLE III
TIME CONSUMPTION AND NUMBER OF TRAINABLE PARAMETERS FOR OFFLINE TRAINING OF KNOWLEDGE ENCODER

Phase	Consumption of Step1		Consumption of Step2	
	Models	Time Cost Parameters	Time Cost Parameters	
Mistral-7B	~ 2h	33.56MB	~ 8h	33.56MB
Qwen-7B	~ 2h	27.53MB	~ 10h	27.53MB
Qwen-14B	~ 4h	47.20MB	~ 18h	47.20MB
Qwen-32B	~ 8h	47.20MB	~ 25h	47.20MB

requirement makes the entire training procedure exceptionally efficient. Consequently, the total offline training time for PriRAG is significantly lower than that required for fine-tuning an entire LLM in practical scenarios, rendering its computational overhead negligible by comparison. We present the associated offline computational costs in Table III. For instance, training the knowledge encoder for the sophisticated Qwen2.5-32B model, a powerful and large LLM, requires approximately 33 hours on just two NVIDIA A800 GPUs. This efficient training regimen highlights PriRAG's practicality for deployment and adaptation even in environments with limited computational resources.

E. Algorithm Analysis

1) *Generalization of Knowledge Encoder*: To thoroughly evaluate the generalization of the PriRAG framework, we deployed pre-trained text embedding modules of diverse architectures and sizes within our knowledge encoder (Fig. 3(d)). The use of different, widely adopted text embedding models consistently yielded strong task performance across all benchmarks. This empirical consistency provides compelling evidence that PriRAG is inherently architecture-agnostic and can be flexibly integrated with a variety of embedding backbones.

2) *Trade-Off Analysis*: We conducted a detailed parameter analysis, specifically comparing PriRAG to the baseline LDP-RAG method under identical privacy budget ϵ settings. The

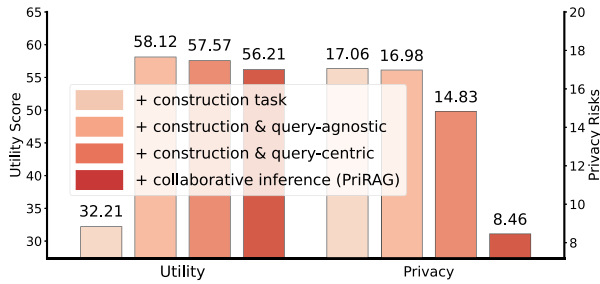


Fig. 5. Ablation study of PriRAG in terms of utility and privacy.

results in Fig. 4 show that PriRAG remains robust across varying noise levels, maintaining strong performance over a wide range of ϵ . In contrast, the LDP-RAG baseline suffers substantial, often prohibitive, performance degradation as ϵ decreases (i.e., under stricter privacy constraints). These findings suggest that the compressed proxy knowledge representations in PriRAG are inherently more compatible with LDP, enabling a more favorable trade-off between utility and privacy.

F. Ablation

Furthermore, to gain deeper insight into the internal mechanisms of PriRAG, we conduct a series of ablation studies to evaluate the impact of different design strategies on both utility and privacy performance. The comprehensive results of these experiments are presented in Fig. 5, revealing three critical insights. First, it can be observed that fine-tuning the knowledge encoder on downstream tasks significantly enhances the ability of the model to interpret and leverage proxy knowledge representations. This adaptation enables the encoder to better align proxy knowledge embeddings with the semantic space of the target LLM, thereby providing a stronger foundation for downstream RAG performance without compromising the inherent privacy guarantees. Second, conditioning the proxy knowledge representations on the user query plays a central role in controlling the utility–privacy trade-off. By selectively emphasizing query-relevant information, this mechanism preserves the most informative signals while naturally limiting the exposure of irrelevant or sensitive content. As a result, it simultaneously enhances task performance and reduces unnecessary information leakage. Comparisons with other pooling methods are provided in Appendix D, available online. Third, introducing DP noise to the proxy knowledge representations substantially reduces privacy leakage with minimal impact on utility. This demonstrates that PriRAG maintains strong robustness under local differential privacy constraints. Importantly, the relatively small utility drop also indicates that the preceding query-centric filtering step helps concentrate informative signals, making the representation more resilient to noise perturbation.

V. CONCLUSION

In this work, we present a privacy-preserving framework (PriRAG) for cloud-based RAG services. We introduce a knowledge encoder that compresses sensitive knowledge into

privacy-preserving proxy representations through a carefully designed mechanism and a collaborative inference method for deployment. The theoretical analysis and empirical evaluation establish formal privacy guarantees. Extensive experiments on seven benchmark datasets show that PriRAG substantially mitigates privacy risks under strong adversarial attacks while consistently matching the inference performance of standard RAG. Further analysis demonstrates that PriRAG not only reduces inference latency but also generalizes well across LLMs of diverse architectures and scales, highlighting its practical applicability in real-world scenarios.

Usage of LLMs: During the writing process, LLMs were utilized to assist with minor linguistic polishing and phrasing refinement to enhance the readability of the manuscript.

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